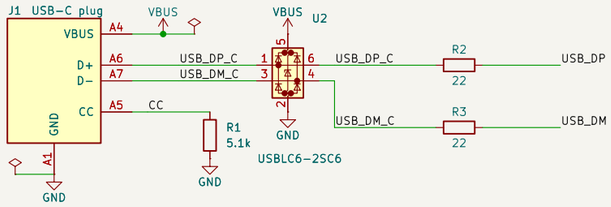
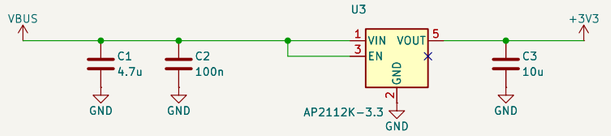
1 USB-C plug and protection



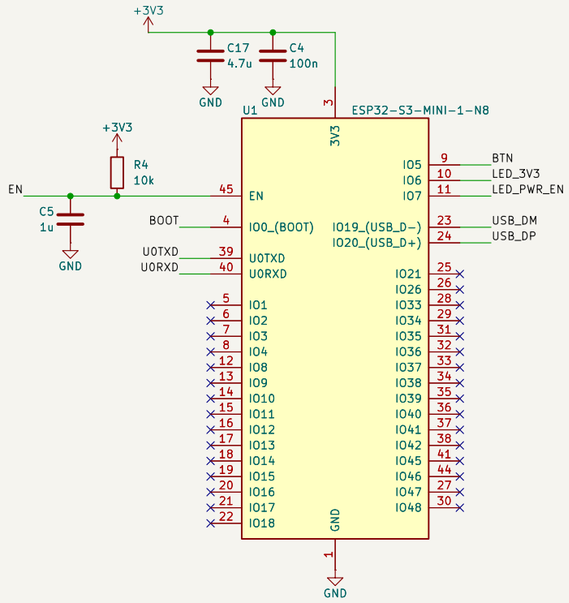
J1 plugs straight into the computer. R1 (5.1 k on CC) asks the host for 5 V.
U2 clamps static on D+, D− and VBUS right behind the plug; R2, R3 (22 R) damp the USB edges.

2 3.3 V supply



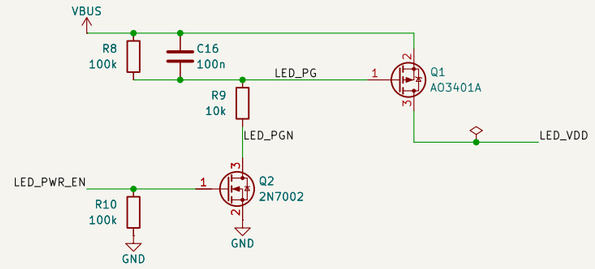
U3: 600 mA LDO for the module (Wi-Fi peaks at about 355 mA). EN tied to VBUS: always on.
C1, C2 at its input, C3 at its output; C17 and C4 sit at the module itself.

3 ESP32-S3 module



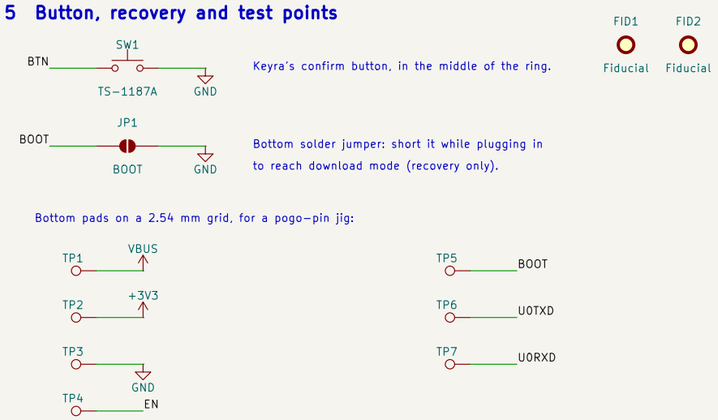
8 MB flash, no PSRAM. The antenna end is the board end: no copper under it.
R4/C5 delay EN at power-up. IO0 keeps the chip's pull-up: JP1 to GND = download mode.
IO19/IO20 are the chip's own USB: keyboard, security key and console.

4 Ring power switch

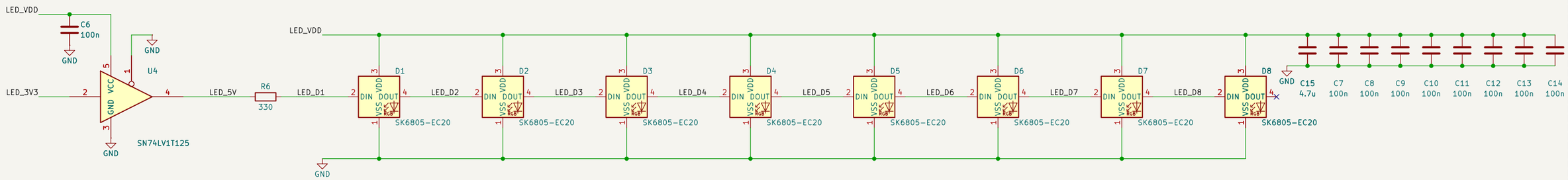


IO7 high: Q2 pulls Q1's gate down and the ring gets 5 V. R10 keeps it off until the firmware runs.
C16 holds Q1 off while VBUS rises at plug-in; R8 || R9 with C16 make the turn-on about 1 ms.

5 Button, recovery and test points



6 Status ring



U4 lifts IO6 (3.3 V) to a 5 V data swing; R6 damps the line. Eight SK6805-EC20 in one chain, all showing the same colour; D8's DOUT stays open.
One 100 nF at each LED (C7 to C14) and 4.7 uF bulk (C15), all on the switched LED_VDD.

Notes

Keyra Stick rev 1.0, Keyra's own board: 4 layers, 0.8 mm, 53.1 x 18.5 mm, ENIG. JLCPCB assembles it from fab/ (BOM, CPL).

Every value, net and position comes from gen/design.py; gen/build.sh redraws this sheet and the board and checks both.

Firmware: the keyra-stick board profile (button IO5, eight LEDs on IO6, their supply on IO7).

Generated by gen/make_sch.py from gen/design.py: edit those, not this file
USB-C stick: ESP32-S3-MINI-1-N8, one button, a ring of eight RGB LEDs
Keyra open hardware (MIT)

Sheet: /
File: keyra-stick.kicad_sch

Title: Keyra Stick

Size: A3	Date: 2026-11-03	**Rev: 1.0**
KiCad E.D.A. 10.0.7		Id: 1/1